Numerical Analysis: Mock Examinations

Computational Methods for Mathematical Problems
Mathematics 101

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January 18, 2026

Contents

Mock Exam 1

MOCK EXAM 1

Numerical Analysis

Duration: 90 minutes | Total: 100 points

Show all work. Partial credit may be awarded for correct reasoning.
Calculators permitted. Round answers to 4 decimal places unless stated otherwise.

Part A: Errors and Approximations (20 points)

Question 1 (10 points)

(a) The exact value of e is 2.71828182.... If we use $\tilde{e} = 2.718$, find the absolute error and relative error (as a percentage). (4 pts)

(b) A computation yields $x = 3.14159$ when the true value is π. How many significant figures are correct? (3 pts)

(c) Explain why computing $\sqrt{x+1} - \sqrt{x}$ for large x may lead to loss of significance. Write an equivalent formula that avoids this problem. (3 pts)

Question 2 (10 points)

(a) How many bisection iterations are needed to achieve an error less than 10^{-4} starting from the interval $[0, 2]$? (5 pts)

(b) If a method has quadratic convergence with $|e_{n+1}| \leq 0.5|e_n|^2$, and the initial error is $|e_0| = 0.1$, estimate $|e_3|$. (5 pts)

Part B: Root Finding Methods (30 points)

Question 3 (15 points)
Consider $f(x) = x^3 - 2x - 5$.

(a) Show that there is a root in the interval $[2, 3]$. (3 pts)

(b) Perform 3 iterations of the bisection method starting from $[2, 3]$. (6 pts)

(c) Perform 2 iterations of Newton-Raphson starting from $x_0 = 2$. (6 pts)

Question 4 (15 points)

(a) Use the secant method with $x_0 = 1$ and $x_1 = 2$ to find x_2 for $f(x) = x^2 - 3$. (5 pts)

(b) For solving $x = \cos(x)$ by fixed-point iteration, verify that the iteration converges by checking $|g'(x)|$ at the approximate root $r \approx 0.739$. (5 pts)

(c) Apply 3 iterations of fixed-point iteration to $x = \cos(x)$ starting from $x_0 = 0.5$. (5 pts)

Part C: Interpolation and Numerical Calculus (25 points)

Question 5 (12 points)
Given the data points: $(0, 1)$, $(1, 3)$, $(2, 7)$.

(a) Construct the Lagrange interpolating polynomial $P_2(x)$. (6 pts)

(b) Use your polynomial to estimate $f(1.5)$. (3 pts)

(c) If the data comes from $f(x) = x^2 + x + 1$, verify your polynomial is correct. (3 pts)

Question 6 (13 points)

(a) Use the central difference formula to approximate $f'(1)$ for $f(x) = \ln(x)$ with $h = 0.1$. Compare with the exact value. (5 pts)

(b) Apply Simpson's rule with $n = 4$ to approximate $\int_0^2 x^2\,dx$. Compare with the exact value. (8 pts)

Part D: Linear Systems and ODEs (25 points)

Question 7 (12 points)
Solve the following system using Gaussian elimination:

$$\begin{cases} x + 2y + z = 9 \\ 2x - y + 3z = 8 \\ 3x + y - z = 3 \end{cases}$$

Question 8 (13 points)
Consider the IVP: $y' = y - t$, $y(0) = 2$.

(a) Apply Euler's method with $h = 0.2$ to find $y(0.4)$. (6 pts)

(b) Apply one step of RK4 with $h = 0.2$ to find $y(0.2)$. (7 pts)

Mock Exam 1 — Solutions

Solution

Question 1

(a) Absolute error: $|2.71828182 - 2.718| = \boxed{0.00028182}$

Relative error: $\frac{0.00028182}{2.71828182} \approx 0.000104 = \boxed{0.0104\%}$

(b) $\pi = 3.14159265...$, approximation $= 3.14159$

Comparing digit by digit: 3.14159 matches 6 significant figures (the 6th digit rounds correctly).

$\boxed{\text{6 significant figures}}$

(c) For large x, both $\sqrt{x+1}$ and $\sqrt{x}$ are nearly equal, so their difference loses precision (catastrophic cancellation).

Rationalized formula:

$$\sqrt{x+1} - \sqrt{x} = \frac{(\sqrt{x+1} - \sqrt{x})(\sqrt{x+1} + \sqrt{x})}{\sqrt{x+1} + \sqrt{x}} = \boxed{\frac{1}{\sqrt{x+1} + \sqrt{x}}}$$

Solution

Question 2

(a) Error after n iterations: $\frac{b-a}{2^{n+1}} < 10^{-4}$

$\frac{2}{2^{n+1}} < 10^{-4} \Rightarrow 2^{n+1} > 2 \times 10^4 \Rightarrow 2^{n+1} > 20000$

$2^{14} = 16384$, $2^{15} = 32768$

So $n + 1 \geq 15$, thus $\boxed{n = 14}$ iterations.

(b) With $|e_{n+1}| \leq 0.5|e_n|^2$:

$$\begin{aligned}
|e_1| &\leq 0.5(0.1)^2 = 0.005 \\
|e_2| &\leq 0.5(0.005)^2 = 0.0000125 \\
|e_3| &\leq 0.5(0.0000125)^2 \approx \boxed{7.8 \times 10^{-11}}
\end{aligned}$$

Solution

Question 3

(a) $f(2) = 8 - 4 - 5 = -1 < 0$

$f(3) = 27 - 6 - 5 = 16 > 0$

Sign change confirms root in $[2, 3]$. ✓

(b) Bisection:

n	a	b	c	$f(c)$
1	2	3	2.5	5.625
2	2	2.5	2.25	1.8906
3	2	2.25	2.125	0.3457

After 3 iterations: $\boxed{r \approx 2.125}$

(c) Newton-Raphson: $f'(x) = 3x^2 - 2$, $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

$x_0 = 2$: $f(2) = -1$, $f'(2) = 10$

$x_1 = 2 - \frac{-1}{10} = \boxed{2.1}$

$x_1 = 2.1$: $f(2.1) = 9.261 - 4.2 - 5 = 0.061$, $f'(2.1) = 13.23 - 2 = 11.23$

$x_2 = 2.1 - \frac{0.061}{11.23} = \boxed{2.0946}$

Solution

Question 4

(a) Secant method: $x_2 = x_1 - f(x_1)\frac{x_1 - x_0}{f(x_1)-f(x_0)}$
$f(1) = 1 - 3 = -2$, $f(2) = 4 - 3 = 1$
$x_2 = 2 - (1)\frac{2-1}{1-(-2)} = 2 - \frac{1}{3} = \boxed{1.6667}$
(b) $g(x) = \cos(x)$, $g'(x) = -\sin(x)$
At $r \approx 0.739$: $|g'(0.739)| = |\sin(0.739)| \approx 0.673 < 1$
Since $|g'(r)| < 1$, the iteration **converges**. ✓
(c) Fixed-point iteration $x_{n+1} = \cos(x_n)$:

$$\begin{aligned} x_0 &= 0.5 \\ x_1 &= \cos(0.5) = 0.8776 \\ x_2 &= \cos(0.8776) = 0.6390 \\ x_3 &= \cos(0.6390) = 0.8027 \end{aligned}$$

$\boxed{x_3 = 0.8027}$ (converging to ≈ 0.739)

Solution

Question 5

(a) Lagrange basis polynomials:

$$\begin{aligned} L_0(x) &= \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{(x-1)(x-2)}{2} \\ L_1(x) &= \frac{(x-0)(x-2)}{(1-0)(1-2)} = -x(x-2) = -x^2 + 2x \\ L_2(x) &= \frac{(x-0)(x-1)}{(2-0)(2-1)} = \frac{x(x-1)}{2} \end{aligned}$$

$P_2(x) = 1 \cdot L_0 + 3 \cdot L_1 + 7 \cdot L_2$
$= \frac{(x-1)(x-2)}{2} + 3(-x^2 + 2x) + 7\frac{x(x-1)}{2}$
$= \frac{x^2-3x+2}{2} - 3x^2 + 6x + \frac{7x^2-7x}{2}$
$= \frac{x^2-3x+2+7x^2-7x}{2} - 3x^2 + 6x = \frac{8x^2-10x+2}{2} - 3x^2 + 6x$
$= 4x^2 - 5x + 1 - 3x^2 + 6x = \boxed{x^2 + x + 1}$
(b) $P_2(1.5) = 2.25 + 1.5 + 1 = \boxed{4.75}$
(c) $f(x) = x^2 + x + 1$: $f(0) = 1$, $f(1) = 3$, $f(2) = 7$ ✓
The polynomial matches exactly.

Solution

Question 6

(a) Central difference: $f'(x) \approx \frac{f(x+h)-f(x-h)}{2h}$
$f'(1) \approx \frac{\ln(1.1)-\ln(0.9)}{0.2} = \frac{0.0953-(-0.1054)}{0.2} = \frac{0.2007}{0.2} = \boxed{1.0034}$
Exact: $f'(x) = \frac{1}{x}$, so $f'(1) = 1$
Error: $|1.0034 - 1| = 0.0034$
(b) Simpson's rule with $n = 4$, $h = 0.5$:
x_i: 0, 0.5, 1, 1.5, 2
$f(x_i) = x_i^2$: 0, 0.25, 1, 2.25, 4

$I \approx \frac{h}{3}[f_0 + 4(f_1 + f_3) + 2f_2 + f_4]$
$= \frac{0.5}{3}[0 + 4(0.25 + 2.25) + 2(1) + 4]$
$= \frac{0.5}{3}[0 + 10 + 2 + 4] = \frac{0.5 \times 16}{3} = \boxed{\dfrac{8}{3} = 2.6667}$
Exact: $\int_0^2 x^2\,dx = \frac{x^3}{3}\Big|_0^2 = \frac{8}{3}$
Simpson's rule gives the exact answer for polynomials of degree ≤ 3.

Solution

Question 7

Augmented matrix: $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 9 \\ 2 & -1 & 3 & 8 \\ 3 & 1 & -1 & 3 \end{array}\right]$
$R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 3R_1$:
$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 9 \\ 0 & -5 & 1 & -10 \\ 0 & -5 & -4 & -24 \end{array}\right]$
$R_3 \leftarrow R_3 - R_2$:
$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 9 \\ 0 & -5 & 1 & -10 \\ 0 & 0 & -5 & -14 \end{array}\right]$
Back substitution: $-5z = -14 \Rightarrow z = \frac{14}{5} = 2.8$
$-5y + 2.8 = -10 \Rightarrow y = 2.56$
$x + 2(2.56) + 2.8 = 9 \Rightarrow x = 1.08$
$\boxed{x = 1.08, \quad y = 2.56, \quad z = 2.8}$

Solution

Question 8

(a) Euler's method: $y_{n+1} = y_n + h \cdot f(t_n, y_n) = y_n + 0.2(y_n - t_n)$
$y_0 = 2$, $t_0 = 0$: $y_1 = 2 + 0.2(2 - 0) = 2 + 0.4 = 2.4$
$y_1 = 2.4$, $t_1 = 0.2$: $y_2 = 2.4 + 0.2(2.4 - 0.2) = 2.4 + 0.44 = 2.84$
$\boxed{y(0.4) \approx 2.84}$
(b) RK4 with $h = 0.2$, $f(t, y) = y - t$, $t_0 = 0$, $y_0 = 2$:
$k_1 = 0.2f(0, 2) = 0.2(2) = 0.4$
$k_2 = 0.2f(0.1, 2.2) = 0.2(2.2 - 0.1) = 0.2(2.1) = 0.42$
$k_3 = 0.2f(0.1, 2.21) = 0.2(2.21 - 0.1) = 0.2(2.11) = 0.422$
$k_4 = 0.2f(0.2, 2.422) = 0.2(2.422 - 0.2) = 0.2(2.222) = 0.4444$
$y_1 = 2 + \frac{1}{6}(0.4 + 2(0.42) + 2(0.422) + 0.4444)$
$= 2 + \frac{1}{6}(0.4 + 0.84 + 0.844 + 0.4444) = 2 + \frac{2.5284}{6}$
$\boxed{y(0.2) \approx 2.4214}$

Mock Exam 2

MOCK EXAM 2

Numerical Analysis

Duration: 90 minutes | Total: 100 points

Show all work. Partial credit may be awarded for correct reasoning.
Calculators permitted. Round answers to 4 decimal places unless stated otherwise.

Part A: Errors and Root Finding (25 points)

Question 1 (10 points)

(a) Given that $\sin(0.1) = 0.0998334...$, if we approximate it as 0.1 (first term of Taylor series), find the absolute and relative errors. (4 pts)

(b) The quadratic formula involves computing $\sqrt{b^2 - 4ac}$. Explain when catastrophic cancellation might occur in calculating $\frac{-b+\sqrt{b^2-4ac}}{2a}$. (3 pts)

(c) What is the order of convergence for the bisection method? For Newton-Raphson? (3 pts)

Question 2 (15 points)
Let $f(x) = e^x - 3x$.

(a) Show there is a root in $[0, 1]$. (3 pts)

(b) Apply 3 iterations of bisection starting from $[0, 1]$. (6 pts)

(c) Set up the Newton-Raphson iteration formula for this function. (3 pts)

(d) Starting from $x_0 = 0.5$, perform 2 Newton-Raphson iterations. (3 pts)

Part B: Interpolation (25 points)

Question 3 (12 points)

(a) Construct the divided difference table for the data:

x	1	2	4
$f(x)$	1	3	9

(6 pts)

(b) Write the Newton interpolating polynomial. (4 pts)

(c) Estimate $f(3)$. (2 pts)

Question 4 (13 points)

(a) State the formula for the interpolation error when using $n + 1$ points. (3 pts)

(b) What is Runge's phenomenon? How can it be mitigated? (4 pts)

(c) Find the Lagrange interpolating polynomial through $(-1, 4)$, $(0, 1)$, $(1, 0)$. (6 pts)

Part C: Numerical Differentiation and Integration (25 points)

Question 5 (12 points)

(a) Write the forward, backward, and central difference formulas for $f'(x)$. Which is most accurate? (4 pts)

(b) Use the second derivative central difference formula to approximate $f''(2)$ for $f(x) = x^3$ with $h = 0.1$. (4 pts)

(c) Explain the trade-off between truncation error and round-off error when choosing h for numerical differentiation. (4 pts)

Question 6 (13 points)

(a) Use the trapezoidal rule with $n = 4$ to approximate $\int_1^3 \frac{1}{x}\,dx$. (6 pts)

(b) What is the exact value of this integral? Calculate the error. (3 pts)

(c) What is the error order of the trapezoidal rule? Of Simpson's rule? (4 pts)

Part D: Linear Systems and ODEs (25 points)

Question 7 (12 points)

(a) Apply 2 iterations of the Jacobi method to:

$$\begin{cases} 5x + y = 11 \\ x + 4y = 18 \end{cases}$$

starting from $(x_0, y_0) = (0, 0)$. (6 pts)

(b) Is this system diagonally dominant? Will the iteration converge? (3 pts)

(c) What is the difference between Jacobi and Gauss-Seidel methods? (3 pts)

Question 8 (13 points)
Consider $y' = -2y$, $y(0) = 1$.

(a) What is the exact solution? (3 pts)

(b) Use Euler's method with $h = 0.25$ to approximate $y(0.5)$. (5 pts)

(c) Compare your Euler approximation with the exact value. (2 pts)

(d) Why is RK4 generally preferred over Euler's method? (3 pts)

Mock Exam 2 — Solutions

Solution

Question 1

(a) Absolute error: $|0.0998334 - 0.1| = \boxed{0.0001666}$

Relative error: $\frac{0.0001666}{0.0998334} \approx 0.00167 = \boxed{0.167\%}$

(b) Catastrophic cancellation occurs when $b > 0$ and $b^2 \gg 4ac$ (discriminant close to b^2). Then $\sqrt{b^2 - 4ac} \approx |b|$, and $-b + \sqrt{b^2 - 4ac}$ involves subtracting nearly equal numbers.

(c) Bisection: **Linear** (order 1), gains ~ 1 bit per iteration.

Newton-Raphson: **Quadratic** (order 2), digits double each iteration.

Solution

Question 2

(a) $f(0) = 1 - 0 = 1 > 0$

$f(1) = e - 3 \approx 2.718 - 3 = -0.282 < 0$

Sign change $\Rightarrow$ root exists in $[0, 1]$. ✓

(b) Bisection:

n	a	b	c	$f(c)$
1	0	1	0.5	0.149
2	0.5	1	0.75	-0.133
3	0.5	0.75	0.625	-0.006

Root $\approx \boxed{0.625}$

(c) $f'(x) = e^x - 3$

$$\boxed{x_{n+1} = x_n - \frac{e^{x_n} - 3x_n}{e^{x_n} - 3}}$$

(d) $x_0 = 0.5$: $f(0.5) = 1.649 - 1.5 = 0.149,\ f'(0.5) = 1.649 - 3 = -1.351$

$x_1 = 0.5 - \frac{0.149}{-1.351} = 0.5 + 0.110 = 0.610$

$x_1 = 0.610$: $f(0.610) = 1.840 - 1.830 = 0.010,\ f'(0.610) = -1.160$

$x_2 = 0.610 - \frac{0.010}{-1.160} = \boxed{0.619}$

Solution

Question 3

(a) Divided difference table:

x	$f[x]$	$f[\cdot,\cdot]$	$f[\cdot,\cdot,\cdot]$
1	1		
		$\frac{3-1}{2-1} = 2$	
2	3		$\frac{2-2}{4-1} = 0$
		$\frac{9-3}{4-2} = 3$	
4	9		

Wait, let me recalculate: $f[x_1, x_2] = \frac{9-3}{4-2} = 3,\ f[x_0, x_1, x_2] = \frac{3-2}{4-1} = \frac{1}{3}$

(b) $P_2(x) = f[x_0] + f[x_0, x_1](x - 1) + f[x_0, x_1, x_2](x - 1)(x - 2)$

$$\boxed{P_2(x) = 1 + 2(x - 1) + \frac{1}{3}(x - 1)(x - 2)}$$

(c) $P_2(3) = 1 + 2(2) + \frac{1}{3}(2)(1) = 1 + 4 + \frac{2}{3} = \boxed{5.667}$

Solution

Question 4

(a) $\boxed{f(x) - P_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^{n}(x - x_i)}$
for some ξ in the interval containing x and all x_i.
(b) **Runge's phenomenon**: High-degree polynomial interpolation with equally spaced points can cause large oscillations near the endpoints.
Mitigation: Use Chebyshev nodes (clustered near endpoints), use lower-degree piecewise polynomials (splines), or use fewer points.
(c) Lagrange polynomial through $(-1, 4)$, $(0, 1)$, $(1, 0)$:
$L_0(x) = \frac{(x-0)(x-1)}{(-1-0)(-1-1)} = \frac{x(x-1)}{2}$
$L_1(x) = \frac{(x+1)(x-1)}{(0+1)(0-1)} = -(x^2 - 1) = 1 - x^2$
$L_2(x) = \frac{(x+1)(x-0)}{(1+1)(1-0)} = \frac{x(x+1)}{2}$
$P_2(x) = 4 \cdot \frac{x(x-1)}{2} + 1 \cdot (1 - x^2) + 0 \cdot \frac{x(x+1)}{2}$
$= 2x^2 - 2x + 1 - x^2 = \boxed{x^2 - 2x + 1} = (x - 1)^2$

Solution

Question 5

(a) Forward: $f'(x) \approx \frac{f(x+h)-f(x)}{h}$, error $O(h)$
Backward: $f'(x) \approx \frac{f(x)-f(x-h)}{h}$, error $O(h)$
Central: $f'(x) \approx \frac{f(x+h)-f(x-h)}{2h}$, error $O(h^2)$ $\leftarrow$ **most accurate**
(b) $f''(x) \approx \frac{f(x+h)-2f(x)+f(x-h)}{h^2}$
$f(1.9) = 6.859$, $f(2) = 8$, $f(2.1) = 9.261$
$f''(2) \approx \frac{9.261-16+6.859}{0.01} = \frac{0.12}{0.01} = \boxed{12}$
Exact: $f''(x) = 6x$, so $f''(2) = 12$ ✓
(c) **Truncation error** decreases as $h \to 0$ (from Taylor series approximation).
Round-off error increases as $h \to 0$ (subtracting nearly equal numbers).
There is an optimal h that balances these two sources of error.

Solution

Question 6

(a) Trapezoidal rule: $h = \frac{3-1}{4} = 0.5$
x: 1, 1.5, 2, 2.5, 3
$f(x) = 1/x$: 1, 0.6667, 0.5, 0.4, 0.3333
$I \approx \frac{0.5}{2}[1 + 2(0.6667 + 0.5 + 0.4) + 0.3333]$
$= 0.25[1 + 3.1334 + 0.3333] = 0.25 \times 4.4667 = \boxed{1.1167}$
(b) Exact: $\int_1^3 \frac{1}{x}\,dx = \ln(3) - \ln(1) = \ln(3) \approx 1.0986$
Error: $|1.1167 - 1.0986| = \boxed{0.0181}$
(c) Trapezoidal rule: $\boxed{O(h^2)}$
Simpson's rule: $\boxed{O(h^4)}$

Solution

Question 7
(a) Rearranged: $x = \frac{11-y}{5}$, $y = \frac{18-x}{4}$
Jacobi iteration:

$$k = 1: \quad x^{(1)} = \frac{11-0}{5} = 2.2, \quad y^{(1)} = \frac{18-0}{4} = 4.5$$
$$k = 2: \quad x^{(2)} = \frac{11-4.5}{5} = 1.3, \quad y^{(2)} = \frac{18-2.2}{4} = 3.95$$

After 2 iterations: $\boxed{x \approx 1.3, \quad y \approx 3.95}$
(b) Diagonal dominance: $|5| > |1|$ and $|4| > |1|$ ✓
Yes, **diagonally dominant**, so iteration will **converge**.
(c) **Jacobi**: Uses values from previous iteration only.
Gauss-Seidel: Uses updated values immediately (faster convergence).

Solution

Question 8
(a) $\frac{dy}{y} = -2\,dt \Rightarrow \ln|y| = -2t + C$
$y(0) = 1 \Rightarrow C = 0$
$\boxed{y = e^{-2t}}$
(b) Euler: $y_{n+1} = y_n + h(-2y_n) = y_n(1-2h) = 0.5y_n$
$y_0 = 1$
$y_1 = 0.5(1) = 0.5$
$y_2 = 0.5(0.5) = 0.25$
$\boxed{y(0.5) \approx 0.25}$
(c) Exact: $y(0.5) = e^{-1} \approx 0.3679$
Euler gives 0.25, error ≈ 0.118 (32% relative error)
(d) RK4 is preferred because:

- Fourth-order accuracy ($O(h^4)$) vs first-order ($O(h)$)
- Much more accurate for same step size
- Better stability properties
- Cost of 4 function evaluations is usually worthwhile

Mock Exam 3

MOCK EXAM 3

Numerical Analysis

Duration: 90 minutes | Total: 100 points

Show all work. Partial credit may be awarded for correct reasoning.
Calculators permitted. Round answers to 4 decimal places unless stated otherwise.

Part A: Comprehensive Root Finding (25 points)

Question 1 (12 points)
The equation $x = 2\sin(x)$ has a nonzero root near $x = 1.9$.

(a) Rewrite as $f(x) = 0$ and verify there is a root in $[1.5, 2]$. (3 pts)

(b) Perform 2 iterations of Newton-Raphson starting from $x_0 = 1.9$. (6 pts)

(c) Rewrite as $x = g(x)$ for fixed-point iteration. Will $g(x) = 2\sin(x)$ converge? (3 pts)

Question 2 (13 points)

(a) Compare bisection, Newton-Raphson, and secant methods in terms of:

- Convergence rate
- Required information (function values, derivatives)
- Reliability

(6 pts)

(b) When would you choose bisection over Newton-Raphson? (3 pts)

(c) What happens to Newton-Raphson near a root of multiplicity 2? (4 pts)

Part B: Interpolation and Approximation (25 points)

Question 3 (13 points)
The following data represents temperature readings:

Time (hr)	0	2	4	6
Temp (°C)	20	26	30	28

(a) Use Lagrange interpolation to estimate the temperature at $t = 3$. (8 pts)

(b) Would you trust this interpolation for $t = 10$? Explain. (2 pts)

(c) What degree is the interpolating polynomial? (3 pts)

Question 4 (12 points)

(a) Construct the Newton divided difference polynomial for $(0, 1)$, $(1, 0)$, $(2, 1)$. (6 pts)

(b) Add the point $(3, 4)$ to your polynomial. What is the advantage of Newton's form for this? (6 pts)

Part C: Numerical Integration (25 points)

Question 5 (13 points)
Approximate $\int_0^\pi \sin(x)\,dx$ using:

(a) The trapezoidal rule with $n = 2$. (4 pts)

(b) Simpson's rule with $n = 2$. (4 pts)

(c) Compare both with the exact value and discuss which is more accurate. (5 pts)

Question 6 (12 points)

(a) Derive the trapezoidal rule by integrating the linear interpolant through $(a, f(a))$ and $(b, f(b))$. (6 pts)

(b) Simpson's rule is exact for polynomials up to what degree? (2 pts)

(c) How does the composite trapezoidal rule differ from the simple trapezoidal rule? (4 pts)

Part D: Systems and ODEs (25 points)

Question 7 (12 points)

(a) Factor $A = LU$ for $A = \begin{pmatrix} 2 & 1 \\ 4 & 5 \end{pmatrix}$. (6 pts)

(b) Use your LU factorization to solve $A\mathbf{x} = \begin{pmatrix} 5 \\ 13 \end{pmatrix}$. (6 pts)

Question 8 (13 points)
For the IVP $y' = t + y$, $y(0) = 1$:

(a) Compute one step of RK2 (midpoint method) with $h = 0.2$. (6 pts)

(b) Compute one step of RK4 with $h = 0.2$. (7 pts)

Mock Exam 3 — Solutions

Solution

Question 1
(a) $f(x) = x - 2\sin(x) = 0$
$f(1.5) = 1.5 - 2\sin(1.5) = 1.5 - 1.995 = -0.495 < 0$
$f(2) = 2 - 2\sin(2) = 2 - 1.819 = 0.181 > 0$
Sign change $\Rightarrow$ root in $[1.5, 2]$. ✓
(b) $f'(x) = 1 - 2\cos(x)$
$x_0 = 1.9$: $f(1.9) = 1.9 - 2\sin(1.9) = 1.9 - 1.892 = 0.008$
$f'(1.9) = 1 - 2\cos(1.9) = 1 - 2(-0.323) = 1.646$
$x_1 = 1.9 - \frac{0.008}{1.646} = 1.895$
$x_1 = 1.895$: $f(1.895) \approx 0.0004$, $f'(1.895) \approx 1.64$
$x_2 = 1.895 - \frac{0.0004}{1.64} = \boxed{1.8955}$
(c) $g(x) = 2\sin(x)$, $g'(x) = 2\cos(x)$
At $r \approx 1.9$: $|g'(1.9)| = |2\cos(1.9)| = 2(0.323) = 0.646 < 1$
Yes, it will **converge** since $|g'(r)| < 1$.

Solution

Question 2
(a) Comparison:

	Bisection	Newton	Secant
Convergence	Linear	Quadratic	≈ 1.618
Requirements	f only	f and f'	f only
Reliability	Always*	May diverge	May diverge

*if sign change exists
(b) Choose bisection when:

- Derivative is expensive or unavailable
- Newton-Raphson diverges (bad initial guess)
- Robustness is more important than speed
- Need guaranteed convergence

(c) At a root of multiplicity 2, Newton-Raphson convergence degrades from quadratic to **linear**. The formula can be modified to $x_{n+1} = x_n - 2\frac{f(x_n)}{f'(x_n)}$ to restore quadratic convergence.

Solution

Question 3
(a) Points: $(0, 20)$, $(2, 26)$, $(4, 30)$, $(6, 28)$
At $t = 3$:
$L_0(3) = \frac{(3-2)(3-4)(3-6)}{(0-2)(0-4)(0-6)} = \frac{(1)(-1)(-3)}{(-2)(-4)(-6)} = \frac{3}{-48} = -0.0625$
$L_1(3) = \frac{(3-0)(3-4)(3-6)}{(2-0)(2-4)(2-6)} = \frac{(3)(-1)(-3)}{(2)(-2)(-4)} = \frac{9}{16} = 0.5625$
$L_2(3) = \frac{(3-0)(3-2)(3-6)}{(4-0)(4-2)(4-6)} = \frac{(3)(1)(-3)}{(4)(2)(-2)} = \frac{-9}{-16} = 0.5625$

$L_3(3) = \frac{(3-0)(3-2)(3-4)}{(6-0)(6-2)(6-4)} = \frac{(3)(1)(-1)}{(6)(4)(2)} = \frac{-3}{48} = -0.0625$
$T(3) = 20(-0.0625) + 26(0.5625) + 30(0.5625) + 28(-0.0625)$
$= -1.25 + 14.625 + 16.875 - 1.75 = \boxed{28.5C}$
(b) **No**. Extrapolation beyond the data range ($t = 10 > 6$) is unreliable. Polynomial interpolation can give erratic results outside the data interval.
(c) With 4 points, the polynomial has degree at most $\boxed{3}$ (cubic).

Solution

Question 4
(a) Divided differences for $(0, 1)$, $(1, 0)$, $(2, 1)$:
$f[x_0] = 1$
$f[x_0, x_1] = \frac{0-1}{1-0} = -1$
$f[x_1, x_2] = \frac{1-0}{2-1} = 1$
$f[x_0, x_1, x_2] = \frac{1-(-1)}{2-0} = 1$
$P_2(x) = 1 + (-1)(x - 0) + 1(x - 0)(x - 1) = 1 - x + x^2 - x = \boxed{x^2 - 2x + 1}$
(b) Adding $(3, 4)$:
$f[x_2, x_3] = \frac{4-1}{3-2} = 3$
$f[x_1, x_2, x_3] = \frac{3-1}{3-1} = 1$
$f[x_0, x_1, x_2, x_3] = \frac{1-1}{3-0} = 0$
$P_3(x) = P_2(x) + 0 \cdot (x)(x - 1)(x - 2) = \boxed{x^2 - 2x + 1}$
Advantage: Newton's form allows adding new points without recalculating the entire polynomial—just compute new divided differences and add a term.

Solution

Question 5
(a) Trapezoidal with $n = 2$: $h = \frac{\pi}{2}$
$I \approx \frac{h}{2}[f(0) + 2f(\pi/2) + f(\pi)] = \frac{\pi/2}{2}[0 + 2(1) + 0] = \frac{\pi}{2} \approx \boxed{1.5708}$
(b) Simpson's with $n = 2$: $h = \frac{\pi}{2}$
$I \approx \frac{h}{3}[f(0) + 4f(\pi/2) + f(\pi)] = \frac{\pi/2}{3}[0 + 4(1) + 0] = \frac{2\pi}{3} \approx \boxed{2.0944}$
(c) Exact: $\int_0^\pi \sin(x)\,dx = [-\cos(x)]_0^\pi = -(-1) - (-1) = 2$
Trapezoidal error: $|1.5708 - 2| = 0.429$ (21% error)
Simpson's error: $|2.0944 - 2| = 0.094$ (4.7% error)
Simpson's is more accurate due to higher order ($O(h^4)$ vs $O(h^2)$).

Solution

Question 6
(a) Linear interpolant through $(a, f(a))$ and $(b, f(b))$:
$P_1(x) = f(a) + \frac{f(b)-f(a)}{b-a}(x - a)$
$\int_a^b P_1(x)\,dx = \int_a^b \left[f(a) + \frac{f(b)-f(a)}{b-a}(x - a)\right] dx$
$= f(a)(b - a) + \frac{f(b)-f(a)}{b-a} \cdot \frac{(b-a)^2}{2}$
$= f(a)(b - a) + \frac{f(b)-f(a)}{2}(b - a)$
$= \frac{b-a}{2}[2f(a) + f(b) - f(a)] = \boxed{\frac{b-a}{2}[f(a) + f(b)]}$

(b) Simpson's rule is exact for polynomials up to degree $\boxed{3}$.
(c) **Simple**: One trapezoid over $[a, b]$.
Composite: Divide $[a, b]$ into n subintervals, apply trapezoidal to each, sum results. More accurate for same interval.

Solution

Question 7

(a) $A = \begin{pmatrix} 2 & 1 \\ 4 & 5 \end{pmatrix}$

$m_{21} = \frac{4}{2} = 2$

$U = \begin{pmatrix} 2 & 1 \\ 0 & 5 - 2(1) \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$

$L = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$

$\boxed{L = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}}$

(b) Solve $L\mathbf{y} = \mathbf{b}$: $\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 13 \end{pmatrix}$

$y_1 = 5$, $2(5) + y_2 = 13 \Rightarrow y_2 = 3$

Solve $U\mathbf{x} = \mathbf{y}$: $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}$

$3x_2 = 3 \Rightarrow x_2 = 1$

$2x_1 + 1 = 5 \Rightarrow x_1 = 2$

$\boxed{x_1 = 2, \quad x_2 = 1}$

Solution

Question 8

$f(t, y) = t + y$, $t_0 = 0$, $y_0 = 1$, $h = 0.2$

(a) RK2 (Midpoint):

$k_1 = hf(0, 1) = 0.2(0 + 1) = 0.2$

$k_2 = hf(0.1, 1.1) = 0.2(0.1 + 1.1) = 0.2(1.2) = 0.24$

$y_1 = 1 + k_2 = \boxed{1.24}$

(b) RK4:

$k_1 = 0.2f(0, 1) = 0.2(1) = 0.2$

$k_2 = 0.2f(0.1, 1.1) = 0.2(1.2) = 0.24$

$k_3 = 0.2f(0.1, 1.12) = 0.2(1.22) = 0.244$

$k_4 = 0.2f(0.2, 1.244) = 0.2(1.444) = 0.2888$

$y_1 = 1 + \frac{1}{6}(0.2 + 2(0.24) + 2(0.244) + 0.2888)$

$= 1 + \frac{1}{6}(1.4568) = 1 + 0.2428 = \boxed{1.2428}$